Rebecca De Rosa

Curriculum Vitae

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Summary

Cybersecurity graduate (MSc, Politecnico di Torino, 2026) working on LLM-based agents for software security. My Master's thesis built a multi-agent LLM system that augments CodeQL static analysis and detects substantially more vulnerabilities than CodeQL alone. As a research assistant at the University of Southern Denmark, I designed and evaluated an agentic vulnerability-detection pipeline with two industry partners, and I am now part of the Applied and Interpretable Machine Learning group at SDU, led by Serkan Ayvaz, as Research Assistant in charge of the LLM development of a RAG-based architecture that aims to support university employees in their activities. I am interested in how to evaluate, verify and monitor the behaviour of LLM agents, and in building the benchmarks and methods that make their failures measurable. Details on each project are on my website, rebeccaderosa.github.io.

Education

2024–2026 **MSc in Cybersecurity**, *Politecnico di Torino*, Turin, Italy
Graduated March 2026. Master's thesis carried out at the University of Southern Denmark (SDU), Sønderborg, as an international exchange student (2025–2026).
Thesis: *A Hybrid Multi-Agent Architecture for Enhancing CodeQL Static Analysis with Large Language Models*
Supervisors: Prof. Adam Alami (SDU), Prof. Danilo Giordano and Prof. Matteo Boffa (Politecnico di Torino)
Three-agent LangGraph pipeline (Analyzer, Suggestor, Creator) that validates CodeQL findings, diagnoses gaps in its rules and drafts new queries; the Analyzer improved CodeQL's F1-score about fourfold (0.43 vs 0.11). [thesis]

2020–2024 **BSc in Computer Engineering**, *Politecnico di Torino*, Turin, Italy

Preprints

2026 P. M. Konrad, T. L. Adam, A. C. H. Merrild, R. Terrenzi, **R. De Rosa**, T. Tanyel, S. Ayvaz. *The Open-Box Fallacy: Why AI Deployment Needs a Calibrated Verification Regime*. arXiv:2605.10601. [arXiv]

Research Experience

Jul 2026– **Research Assistant, Applied and Interpretable Machine Learning Group**, *University of Southern Denmark (SDU)*, Sønderborg, Denmark
Designed and implemented a RAG-based architecture that aims to support university employees: OCR (PaddleOCR), table- and paragraph-aware chunking, vector search, and LLM answers with the source passages highlighted for review.

Jan–Jun 2026 **Research Assistant**, *University of Southern Denmark (SDU)*, Sønderborg, Denmark
Beyond Patching – Software Vulnerability Detection: Orchestrating Static Analysis and LLM Reasoning through an Agentic Architecture, with Umbraco and Unik (project led by Prof. Adam Alami). Designed and developed the agentic architecture: LLM agents detect vulnerabilities, explain how they could be exploited and propose automatically tested fixes. Uncovered a likely exploitable flaw in an open-source library.

Teaching

Jul 2026– **Teaching Assistant, Web Development**, *University of Southern Denmark (SDU)*, Sønderborg, Denmark
Bachelor-level course. Lead the exercise sessions, working through PHP and Laravel exercises with students.

2020–2024 **Private Tutor**, Turin, Italy
Physics, Mathematics, Chemistry and Computer Science for middle and high school students.

Technical Skills

LLMs & Agents	LangChain, LangGraph, single- and multi-agent design, closed and open-source models (GPT, Claude, DeepSeek, Qwen), design and implementation of RAG-based architectures, prompt engineering
ML	PyTorch
Security	Static analysis (CodeQL), vulnerability detection
Programming	Python, Java, C, C++, PHP (Laravel); Linux

Languages

Italian	Native
English	Advanced (C1)

Volunteering & Interests

Volunteering	Independent volunteer at the Lea Mwana orphanage in Malindi, Kenya (3 weeks, 2024): organised activities for the children and helped them with their homework.
Training	Three-day retreat with Davide Malaguti (Dreams, 2024) on leadership, proactivity and personal growth.
Work	Alongside my studies: Cambridge English exam invigilator, event staff, and pizza delivery in Denmark.
Interests	Volleyball and kickboxing (one amateur fight).